

E054

# Foundations of Project Management

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

|                             |                                           |                        |                                  |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>_____ | [ ] sure [ ] unsure [ ]<br>guess |
|-----------------------------|-------------------------------------------|------------------------|----------------------------------|

# Items

1.

A regional hospital with 400 beds processes about 2,000 insurance claims each week using the same steps. Leadership also plans to install a new claims system by June. The billing manager has been there 20 years. Which of these is a project?

- A. Both, because both are important to the hospital.
- B. Neither; hospitals do not run projects.
- C. Installing the new claims system by June.
- D. Processing the 2,000 weekly claims.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2.

Midway through building a customer portal for a credit union with a new logo, the sponsor adds two major features but keeps the launch date and budget fixed. The team is already fully booked. What is the most likely result if nothing else changes?

- A. The features can be added with no impact.
- B. The budget will fall.
- C. The new logo will need to be redesigned.
- D. Quality is likely to suffer, so the PM should present trade-offs such as more time, more budget, or less scope.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

At a manufacturer with plants in two states, a project manager must get approval from each department head before using their staff, and team members report to their department heads. The PM has a part-time role and a small office. What organizational structure is this most likely?

- A. Functional
- B. Strong matrix
- C. Virtual
- D. Projectized

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4.

A city is building a new bridge. Engineering specs are fixed, permits require an approved design, and the contractor has built 30 bridges. The city council meets on Tuesdays. Which approach fits best?

- A. Adaptive, with two-week design sprints on the bridge structure.
- B. Predictive (plan-driven)
- C. Adaptive (agile), since agile is always better.
- D. No formal approach, because the contractor is experienced.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5. [TRANSFER — reversed stem / different variable]

A startup is building a mobile app where users' needs are unclear and will be learned from early releases. Which approach fits best?

- A. Adaptive (agile)
- B. Predictive, with all requirements fixed up front.
- C. Neither; skip planning entirely.
- D. Predictive, because apps are software.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A school district is closing two libraries to fund a new digital learning center. The project team has met with the superintendent and IT staff. The district recently repainted its buses. Parents and librarians have not been contacted, and the plan is due in three weeks. What should the PM do?

- A. Engage only supporters to keep things moving.
- B. Delay the plan by three more weeks.
- C. Wait until the plan is final and then announce it.
- D. Identify and engage parents and librarians now, including likely opponents, and plan how to involve them.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7.

A marketing project team of 8 people, spread across two floors of a new office tower, meets weekly. How many communication channels exist on the team?

- A. 16
- B. 56
- C. 8
- D. 28

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8. [TRANSFER — reversed stem / different variable]

A team grows from 5 to 8 people. How many communication channels are added?

- A. 3
- B. 18
- C. 6
- D. 28

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

9.

A software rollout has activity A (3 days), then B (5 days) and C (7 days) in parallel after A, then D (2 days) after both B and C. The vendor's office is in Toronto. What is the critical path and project duration?

- A. A-B-D, 10 days
- B. A-C-D, 9 days
- C. A-C-D, 12 days
- D. A-B-C-D, 17 days

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

10. [TRANSFER — reversed stem / different variable]

Using the same network (A 3, B 5, C 7, D 2; B and C in parallel), how much total float does activity B have?

- A. 5 days
- B. 7 days
- C. 2 days
- D. 0 days

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A \$100,000 website project, managed by a PM who just earned a certification, was planned to be 50% complete by now. It is 40% complete, and \$45,000 has been spent. What is the CPI, and what does it mean?

- A.** 1.13; under budget
- B.** 0.89; over budget
- C.** 0.80; behind schedule
- D.** 0.89; under budget

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

**12.** [TRANSFER — reversed stem / different variable]

A project has a CPI of 1.25 and actual cost of \$80,000. What is its earned value?

- A.** \$100,000
- B.** \$64,000
- C.** \$80,000
- D.** \$20,000

|                             |                                           |                       |                                                                                                 |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|
| <b>Target word</b><br>_____ | <b>The one fact that decides</b><br>_____ | <b>Answer</b><br>____ | <input type="checkbox"/> sure <input type="checkbox"/> unsure <input type="checkbox"/><br>guess |
|-----------------------------|-------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

A regional hospital ~~with 400 beds~~ processes about 2,000 insurance claims each week using the same steps. Leadership also plans to install a new claims system by June. ~~The billing manager has been there 20 years.~~ Which of these is a project?

**Target word:** project vs. operations **Decides:** Temporary with a unique result.

|   |                   |                                                  |
|---|-------------------|--------------------------------------------------|
| A | Adjacent category | Importance does not define a project.            |
| B | Absolute language | Any organization can run projects.               |
| C | <b>Correct</b>    | Temporary, with a defined end and unique result. |
| D | Confusion set     | Ongoing, repetitive operations.                  |

## 2. Answer D Concept 2

Midway through building a customer portal for a credit union ~~with a new logo~~, the sponsor adds two major features but keeps the launch date and budget fixed. The team is already fully booked. What is the most likely result if nothing else changes?

**Target word:** constraint trade-off **Decides:** More scope with fixed time, cost, and resources.

|   |                    |                                              |
|---|--------------------|----------------------------------------------|
| A | Absolute language  | Ignores the linked constraints.              |
| B | Inverted operation | More scope raises cost pressure.             |
| C | Camouflage         | Unrelated to the trade-off.                  |
| D | <b>Correct</b>     | Constraints are linked; something must give. |

## 3. Answer A Concept 3

At a manufacturer ~~with plants in two states~~, a project manager must get approval from each department head before using their staff, and team members report to their department heads. The PM has a part-time role ~~and a small office~~. What organizational structure is this most likely?

**Target word:** structure **Decides:** Low PM authority, staff report to functional managers.

|   |                    |                                                      |
|---|--------------------|------------------------------------------------------|
| A | <b>Correct</b>     | Functional managers hold authority; the PM's is low. |
| B | Adjacent category  | A full-time PM would share significant authority.    |
| C | Confusion set      | Describes location, not authority.                   |
| D | Inverted operation | The PM would control staff and budget.               |

## 4. Answer B Concept 4

A city is building a new bridge. Engineering specs are fixed, permits require an approved design, and ~~the contractor has built 30 bridges. The city council meets on Tuesdays.~~ Which approach fits best?

**Target word:** predictive vs. adaptive **Decides:** Stable, fixed requirements.

|   |                    |                                                                  |
|---|--------------------|------------------------------------------------------------------|
| A | Inverted operation | Fixed specs and permits leave little to iterate.                 |
| B | <b>Correct</b>     | Stable requirements and approved designs suit up-front planning. |
| C | Absolute language  | Agile suits uncertainty, which is absent here.                   |
| D | Camouflage         | Experience does not replace an approach.                         |

### 5. Answer A Concept 4 • transfer

A startup is building a mobile app where users' needs are unclear and will be learned from early releases. Which approach fits best?

**Target word:** opposite case **Decides:** Evolving requirements.

|   |                    |                                                               |
|---|--------------------|---------------------------------------------------------------|
| A | <b>Correct</b>     | Iterations with feedback suit uncertain needs.                |
| B | Inverted operation | Unclear needs cannot be fixed up front.                       |
| C | Dropped element    | Agile still plans, in shorter cycles.                         |
| D | Confusion set      | The product type does not decide; requirement stability does. |

### 6. Answer D Concept 5

A school district is closing two libraries to fund a new digital learning center. The project team has met with the superintendent and IT staff. ~~The district recently repainted its buses.~~ Parents and librarians have not been contacted, and the plan is due in three weeks. What should the PM do?

**Target word:** stakeholder identification **Decides:** Affected groups not yet engaged.

|   |                    |                                                                   |
|---|--------------------|-------------------------------------------------------------------|
| A | Dropped element    | Leaves out opponents who can derail the project.                  |
| B | Wrong stem number  | The timeline does not replace engagement.                         |
| C | Inverted operation | Late engagement breeds opposition.                                |
| D | <b>Correct</b>     | Early engagement of all affected groups reduces later resistance. |

### 7. Answer D Concept 6

A marketing project team of 8 people, ~~spread across two floors of a new office tower,~~ meets weekly. How many communication channels exist on the team?

**Target word:** channels **Decides:**  $8 \times 7 / 2$ .

|   |                   |                             |
|---|-------------------|-----------------------------|
| A | Confusion set     | $2n$ assumes linear growth. |
| B | Dropped element   | Leaves out dividing by 2.   |
| C | Wrong stem number | The team size.              |
| D | <b>Correct</b>    | $8 \times 7 / 2 = 28$ .     |

### 8. Answer B Concept 6 • transfer

A team grows from 5 to 8 people. How many communication channels are added?

**Target word:** change in channels **Decides:**  $28 - 10$ .

|   |                   |                                                  |
|---|-------------------|--------------------------------------------------|
| A | Wrong stem number | The number of people added.                      |
| B | <b>Correct</b>    | 5 people = 10 channels; 8 = 28; $28 - 10 = 18$ . |
| C | Confusion set     | Assumes 2 channels per new person.               |
| D | Dropped element   | Total after growth, not the increase.            |

## 9. Answer C Concept 7

A software rollout has activity A (3 days), then B (5 days) and C (7 days) in parallel after A, then D (2 days) after both B and C. ~~The vendor's office is in Toronto.~~ What is the critical path and project duration?

**Target word:** critical path **Decides:** A-C-D = 12 vs. A-B-D = 10.

|          |                    |                                                         |
|----------|--------------------|---------------------------------------------------------|
| A        | Inverted operation | The shortest path, not the longest.                     |
| B        | Dropped element    | Right path, but leaves out activity A's 3 days (7 + 2). |
| <b>C</b> | <b>Correct</b>     | Longest path: 3 + 7 + 2 = 12.                           |
| D        | Confusion set      | Adds parallel activities as if sequential.              |

## 10. Answer C Concept 7 • transfer

Using the same network (A 3, B 5, C 7, D 2; B and C in parallel), how much total float does activity B have?

**Target word:** solve for float **Decides:** 12 – 10.

|          |                    |                                                           |
|----------|--------------------|-----------------------------------------------------------|
| A        | Wrong stem number  | B's duration.                                             |
| B        | Inverted operation | C's duration, not the path difference.                    |
| <b>C</b> | <b>Correct</b>     | Critical path 12 – B's path 10 = 2.                       |
| D        | Confusion set      | Zero float is for critical activities; B is not critical. |

## 11. Answer B Concept 8

A \$100,000 website project, ~~managed by a PM who just earned a certification,~~ was planned to be 50% complete by now. It is 40% complete, and \$45,000 has been spent. What is the CPI, and what does it mean?

**Target word:** CPI **Decides:** EV = \$40,000; AC = \$45,000.

|          |                    |                                                             |
|----------|--------------------|-------------------------------------------------------------|
| A        | Inverted operation | Divides AC by EV.                                           |
| <b>B</b> | <b>Correct</b>     | $\$40,000 / \$45,000 = 0.89$ ; below 1.0 means over budget. |
| C        | Confusion set      | That is SPI = $\$40,000 / \$50,000$ .                       |
| D        | Adjacent category  | Right number, wrong reading.                                |

## 12. Answer A Concept 8 • transfer

A project has a CPI of 1.25 and actual cost of \$80,000. What is its earned value?

**Target word:** solve for EV **Decides:** EV = CPI × AC.

|          |                    |                                                  |
|----------|--------------------|--------------------------------------------------|
| <b>A</b> | <b>Correct</b>     | $1.25 \times \$80,000 = \$100,000$ .             |
| B        | Inverted operation | $\$80,000 / 1.25$ divides instead of multiplies. |
| C        | Wrong stem number  | The actual cost.                                 |
| D        | Dropped element    | Only the gap between EV and AC.                  |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 6             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 5             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 3             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 9             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 3             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 8             |             |
| Camouflage         | Answers a stem fact that was never the question.             | 2             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                          | Rule                                                                                                                                                                                 | Trap                                                                         | Tell                                                         |
|---|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|--------------------------------------------------------------|
| 1 | <b>Projects vs. operations</b>                   | A project is a temporary effort to create a unique product, service, or result; operations are ongoing and repetitive.                                                               | Calling any big or important piece of work a project.                        | Whether the work has a defined end and a unique output.      |
| 2 | <b>Competing constraints</b>                     | Scope, schedule, cost, quality, and resources are linked; changing one forces a trade-off in the others.                                                                             | Adding scope without adjusting time, budget, or resources.                   | A new request with a fixed deadline and budget.              |
| 3 | <b>Organizational structure and PM authority</b> | Project manager authority is low in functional organizations, shared in matrix organizations (weak to strong), and high in projectized organizations.                                | Assuming the PM controls resources in a functional organization.             | Who team members report to and who controls the budget.      |
| 4 | <b>Choosing predictive vs. adaptive</b>          | Predictive (plan-driven) fits stable, well-understood requirements; adaptive (agile) fits uncertain or evolving requirements needing frequent feedback.                              | Choosing agile for everything or waterfall by habit.                         | How stable and clear the requirements are.                   |
| 5 | <b>Identify and engage stakeholders early</b>    | Stakeholders, including those who may oppose the project, are identified early and engaged according to their interest and influence throughout the life cycle.                      | Engaging only supporters or waiting until launch to involve affected groups. | A group affected by the project that has not been consulted. |
| 6 | <b>Communication channels grow fast</b>          | Channels = $n(n - 1) / 2$ for $n$ people, so each added person adds more paths than the last.                                                                                        | Assuming channels grow in step with team size ( $n$ or $2n$ ).               | A team size and a question about communication paths.        |
| 7 | <b>Critical path and float</b>                   | The critical path is the longest path through the network, sets the shortest possible duration, and has zero total float; float on other paths = critical path length – path length. | Picking the shortest path or the path with the most activities.              | A network of activities with durations and dependencies.     |
| 8 | <b>Earned value indices</b>                      | $CPI = EV / AC$ and $SPI = EV / PV$ ; values below 1.0 mean over budget or behind schedule.                                                                                          | Dividing AC by EV, or reading a CPI below 1 as under budget.                 | Planned value, earned value, and actual cost figures.        |

**Course scope (public description):** Foundations of Project Management provide learners with a comprehensive understanding of core project management principles and practices, emphasizing the alignment of project management processes with organizational goals and structures throughout the project life cycle. Learners will explore the roles and responsibilities of project stakeholders and their impact on project success, learning effective stakeholder engagement strategies and communication techniques. Additionally, learners will develop the skills to determine appropriate project management approaches, tools, and practices based on project characteristics, enabling them to make informed decisions and drive successful project outcomes.